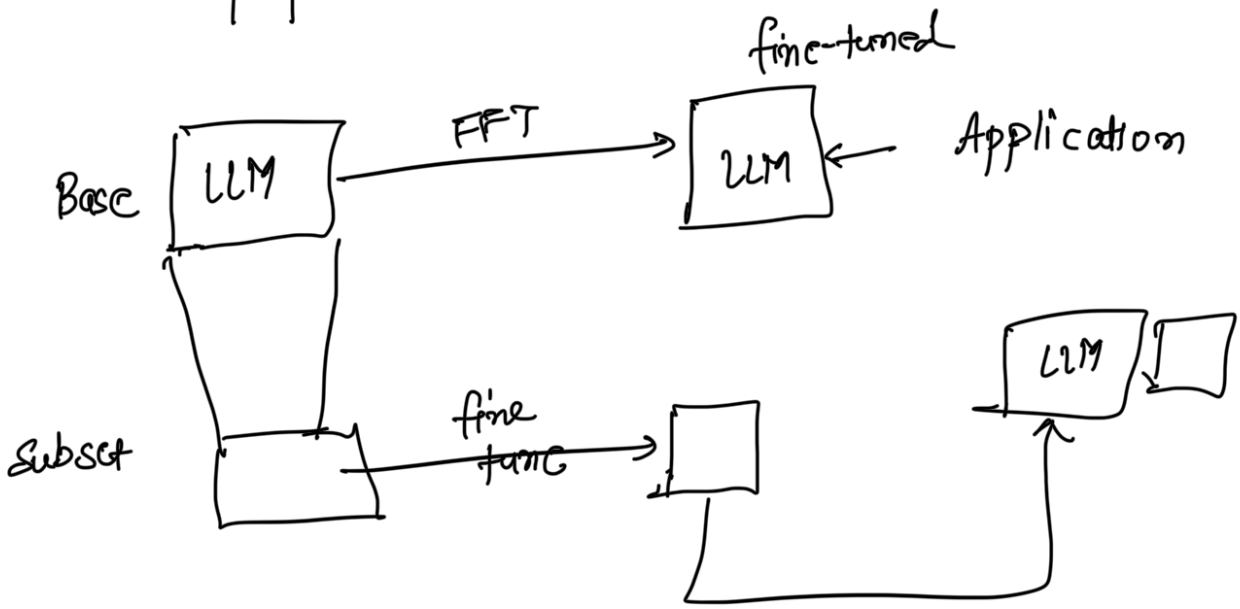
14/3/26

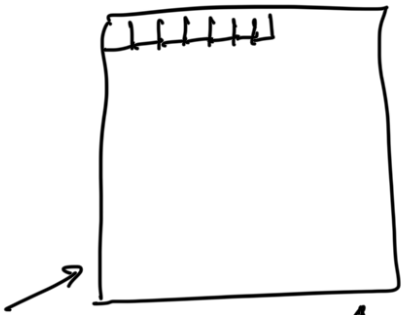
LoRA and QLoRA

Low-Rank Adaptation

↳ Benefit → updating only a small number of parameters



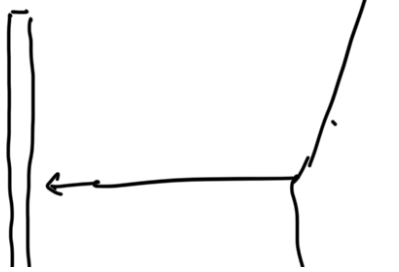
20X20 Matrix



→ 400

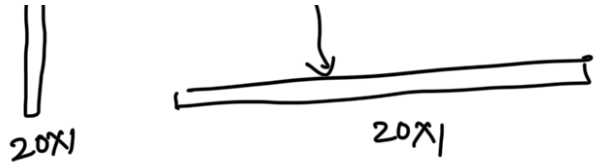
20 X20

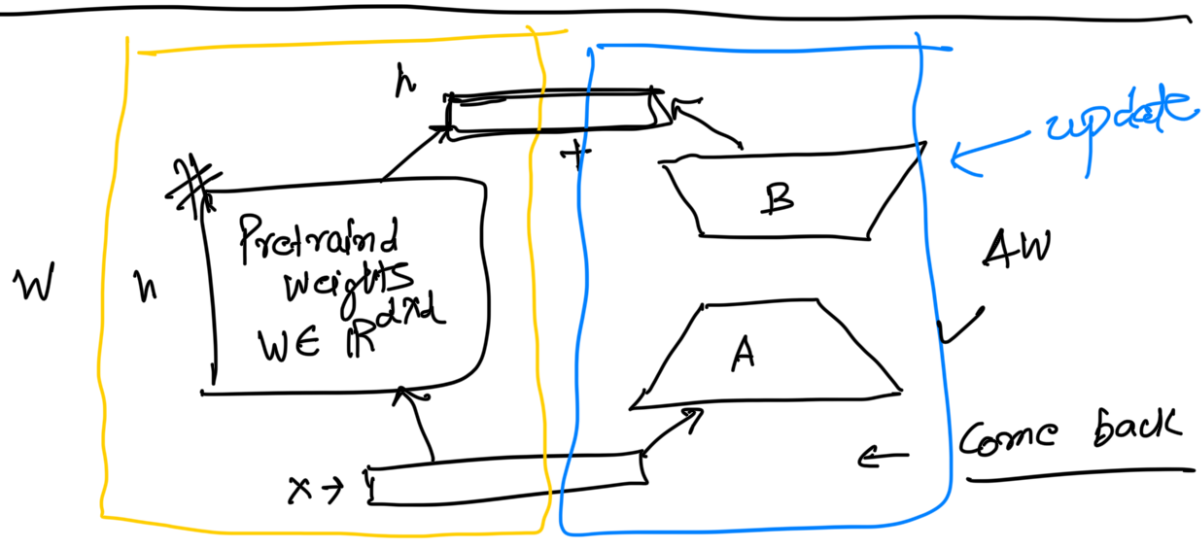
* low-rank weight matrix



← 20+20

40 params





$P_\phi(y|x)$ → Summarize text
→ translation

Task $Z = \{(x_i, y_i)\}_{i=1,\dots,N}$

NL 2SQL → x_i → natural language query
y_i → SQL Command

ϕ_0 → Initial weights
↓
$\phi_0 + \Delta\phi$

fine tuning

GPT-3] 175 Billion parameters

LoRA → 0.01 % of the parameters

Aghajanyan et. al.

Intrinsic dimensionality Explains the Effectiveness of Language Mode fine-tuning

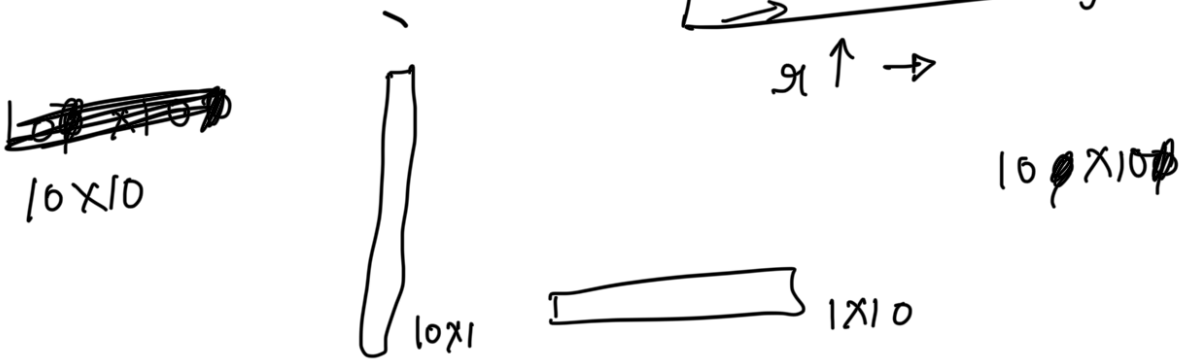
pre-trained weight matrix → $W_0 \in \mathbb{R}^{d \times k}$

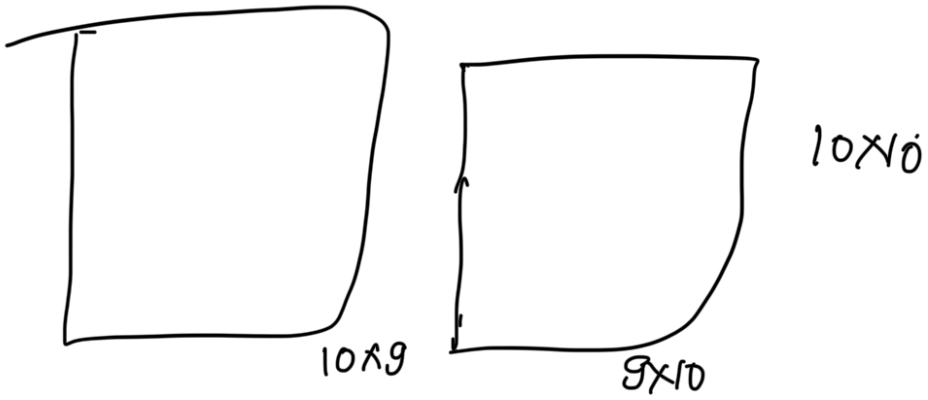
$$W_0 + \underbrace{\Delta W}_{\text{update}} = W_0 + \underbrace{BA}_{\text{low rank - decomposition}}$$

$\Delta W \in \mathbb{R}^{d \times k}$ $\quad B \in \mathbb{R}^{d \times r}$ $\quad A \in \mathbb{R}^{r \times k}$

$$\boxed{r \ll \min(d, k)}$$

$r \uparrow \rightarrow$





$h = W_0 x$ **

$h = (W_0 + \boxed{\Delta W})\, x$

zero

$h = W_0 x + \Delta W x$

$= W_0 x + \Delta W x$

$$\boxed{h = W_0 x + AB x}$$

random gaussian intiallization

$x \uparrow \longrightarrow$ LORA $\longrightarrow$ FT the complete model

$$(W_0 + BA)\, x$$

English $\rightarrow$ Java $\quad W_0 + \boxed{B_1 A_1}$

English $\rightarrow$ Python $\quad W_0 + \boxed{B_2 A_2}$

GPT-3 - 175 B parameters

0.01 % $\rightarrow$ 0.0175 B parameters

QLoRA $\rightarrow$ Efficient finetuning of Quantized LLMs

Quantization is process of discretization of an input from a higher level representation to a lower level representation.

Number → 32-bit floating representation
→ 8-bit integer representation

---

* 32-bit number (floating point) → X^{FP32} → FB → Int 8

(Int 8 tensor) [-127, 127]

$$\text{round}\left[\frac{X^{FP32}}{\text{abs max}[X^{FP32}]} \cdot 127\right] \rightarrow X^{Int8}$$

$$\text{round}\left[\frac{127}{\text{abs max}[X^{FP32}]} \cdot X^{FP32}\right] = X^{Int8}$$

$$\| $$

$$\text{round}\left(C^{FB32} \cdot X^{FP32}\right)$$

$$\text{dequant}\left(C^{FB32}, X^{Int8}\right) = \frac{X^{Int8}}{C^{FB32}} = X^{FB32}$$

---

pi - FL 32, 16

Attempt to represent pi

3.1415927 → 3.141

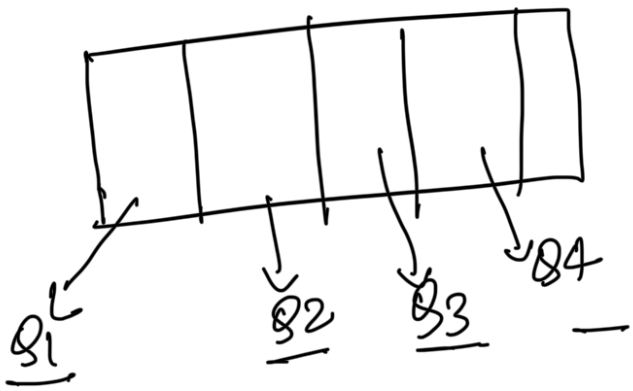
3.1416123 → 3.141

3.1410121 → 3.141

3.1412122 → 3.141

Disadvantage

BLOCK-WISE QUANTIZATION



$X \in \mathbb{R}^{b \times L}$ → n blocks of size B

Block-wise quantization

Low-rank Adapters : → Reducing the memory

7B LLaMA → 0.2% weights of the original model

Gradients (567 MB) ← 26 MB

5048 MB of memory